

Program : Diploma in Cyber Forensics and Information Security	
Course Code : 6289B	Course Title: Mobile Application Security Lab
Semester : 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Design and build secure applications on android and IOS
- Test and protect against security threats

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic Knowledge of Linux OS		System Administration Lab	3

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the Android application threat landscape	9	Applying
CO2	Understand the iOS application threat landscape	9	Applying
CO3	Perform penetration testing on android and iOS mobile apps	11	Applying
CO4	Identify vulnerabilities and exploit them	13	Applying
	Lab Test	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2		3		1	2
CO2	3		2				
CO3	3				1		2
CO4	3		2			2	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Understand the Android application threat landscape		
M1.01	Android Emulator, ADB and Database Isolation	3	Applying
M1.02	Build your own malware app and steal other app files	3	Applying
M1.03	Recovering protected secrets, Application patching	3	Applying
CO2	Understand the iOS application threat landscape		
M2.01	Setting-up and Exploring the iOS environment	3	Applying
M2.02	Binary Static Analysis manual and automated	3	Applying
M2.03	Catch and manipulate application's traffic	3	Applying
	Lab Test – I	1.5	Applying
CO3	Perform penetration testing on android and iOS mobile apps		
M3.01	Invoking Internal Activities Using Malicious Intents	3	Applying
M3.02	Objective C and runtime manipulation using iNalyzer and Cycrypt iOS	4	Applying
M3.03	Parameter Manipulation Using a Proxy	2	Applying

M3.04	iNalyzer Storage Snapshot	2	Applying
C04	Identify vulnerabilities and exploit them		
M4.01	Monitoring network connectivity	2	Applying
M4.02	Analyzing logs using logcat	2	Applying
M4.03	Memory dumps and analysis	2	Applying
M4.04	Smali Debugging	2	Applying
	Open ended experiments	5	
	Lab Test – II	1.5	

Sample Open Ended Experiments

(Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

- Source Code Encryption
- Penetration Tests - Perform a Thorough QA & Security Check
- Secure the Data-in-transit
- Use the Latest Cryptography Techniques
- High-level Authentication

Text / Reference

T/R	Book Title/Author
T1	Mobile Application Security by David Thiel, Chris Clark, Himanshu Dwivedi
T2	Mobile Application Security McGraw-Hill, Inc. Professional Book Group 11 West 19th Street New York, NY

Online Resources

Sl.No	Website Link
1	https://www.elearnsecurity.com/maspt